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Digitalising Social Protection: Better Targeting and Delivery

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BIO

Social assistance in Indonesia misses a large share of the households it is meant for. This asks three questions: how large the gap is, where it comes from, and which parts of it digitalisation can actually close.

THE PROBLEM

We cannot see who is poor. We only see signs.

A hunter cannot see the animal in the tree. He can only see the leaves move. Targeting works the same way: welfare itself is never observed, only proxies for it.

Who decides

Tree A hides the **musang** — the household that qualifies. Tree B holds **birds** — households that look the same from outside.

Exclusion error — the musang escapes: a household that qualifies gets nothing

Inclusion error — a bird is hit: a household that does not qualify receives the benefit

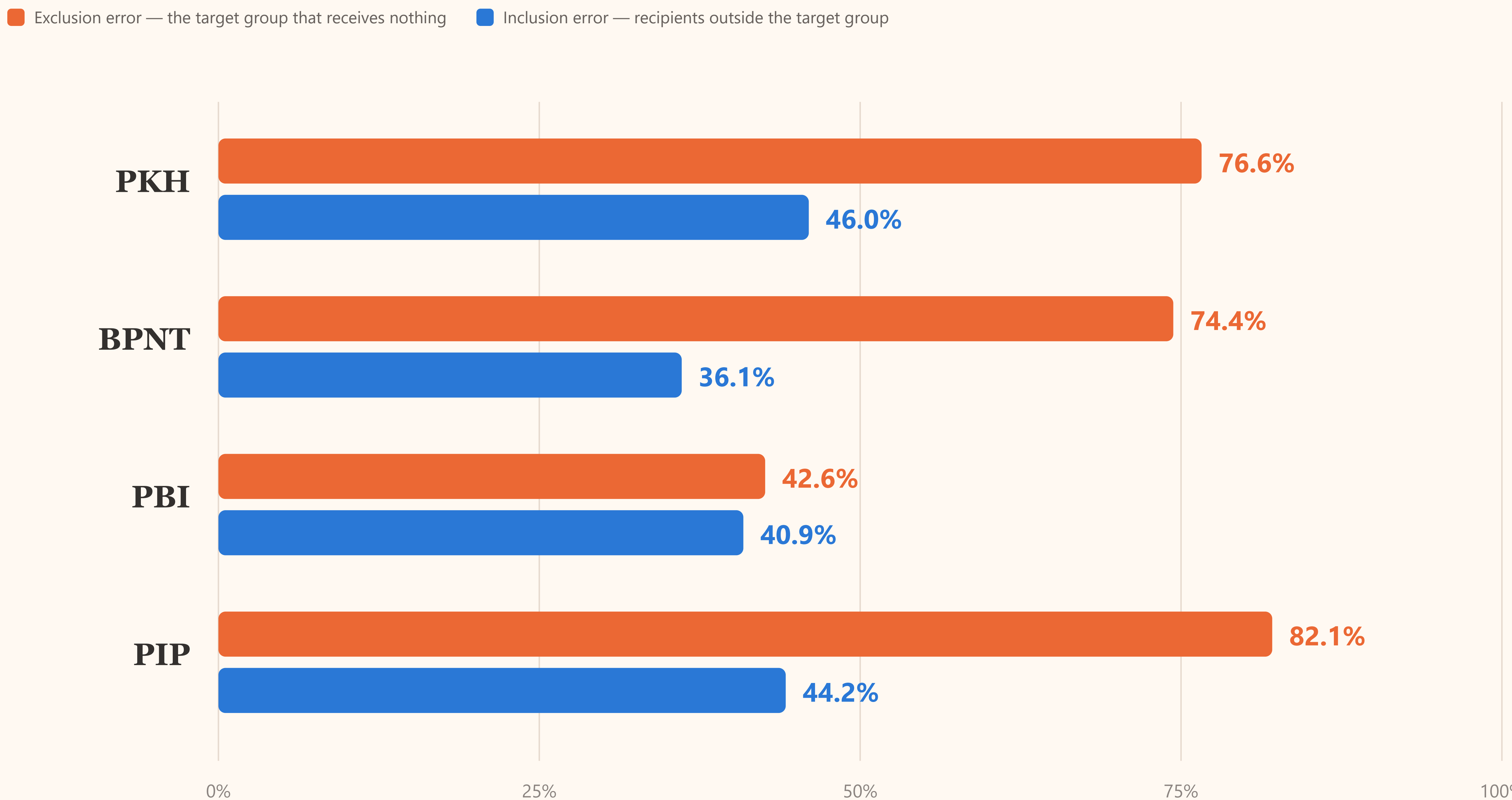
I How well does it reach people?

Four programmes, measured against the households and people they are meant to serve.

FIGURE 1

How much of the target group is missed

Susenas March 2025. Exclusion error is the share of the target group that receives nothing. Inclusion error is the share of recipients who fall outside the target group.



Move the PKH cut-off to deciles 1–3 and inclusion error becomes **57.7%**; move it to 1–5 and it becomes **35.7%**.
Nothing about the programme changed — only where the line is drawn.

FIGURE 2

Three years: two programmes improving, one not

Exclusion error, March 2023 to March 2025, on the same definitions. PBI and PIP reach more of their target group each year. PKH does not.

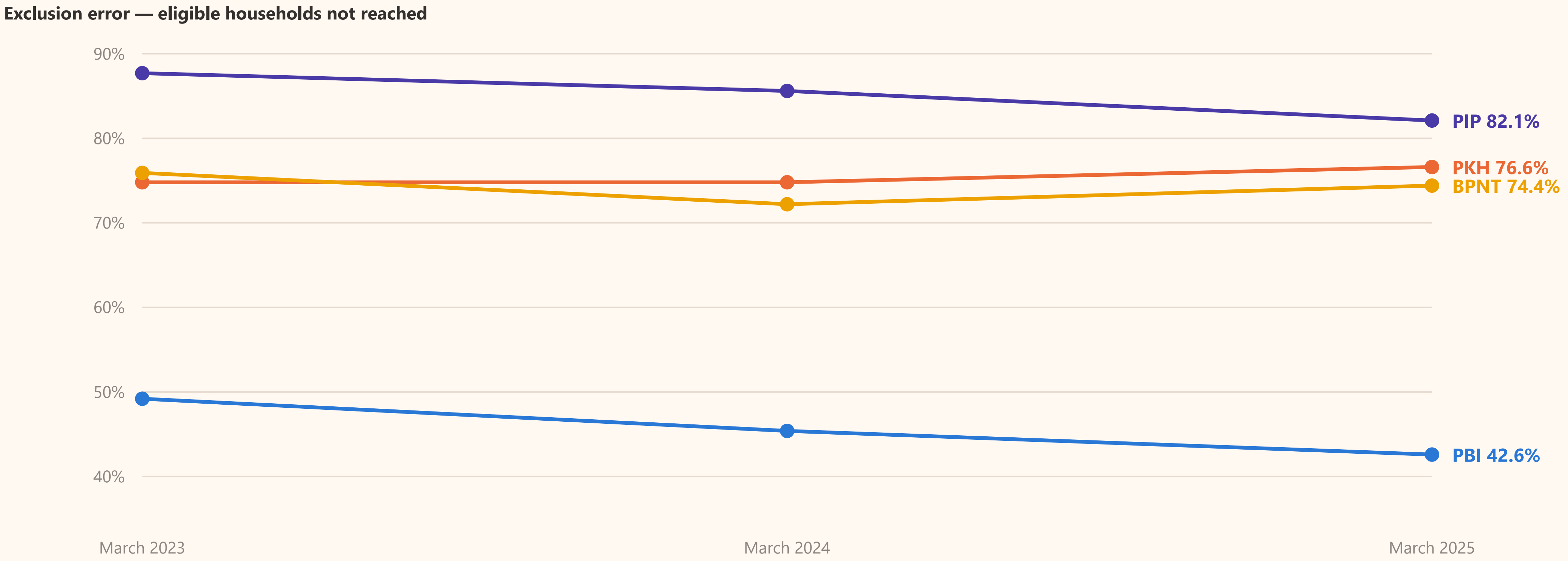


FIGURE 3

The ranking works. The coverage does not.

Share of households or people receiving each programme, by welfare decile. Coverage falls steadily from the poorest decile to the richest in every programme, so the ranking is pointing the right way.

Share receiving the programme

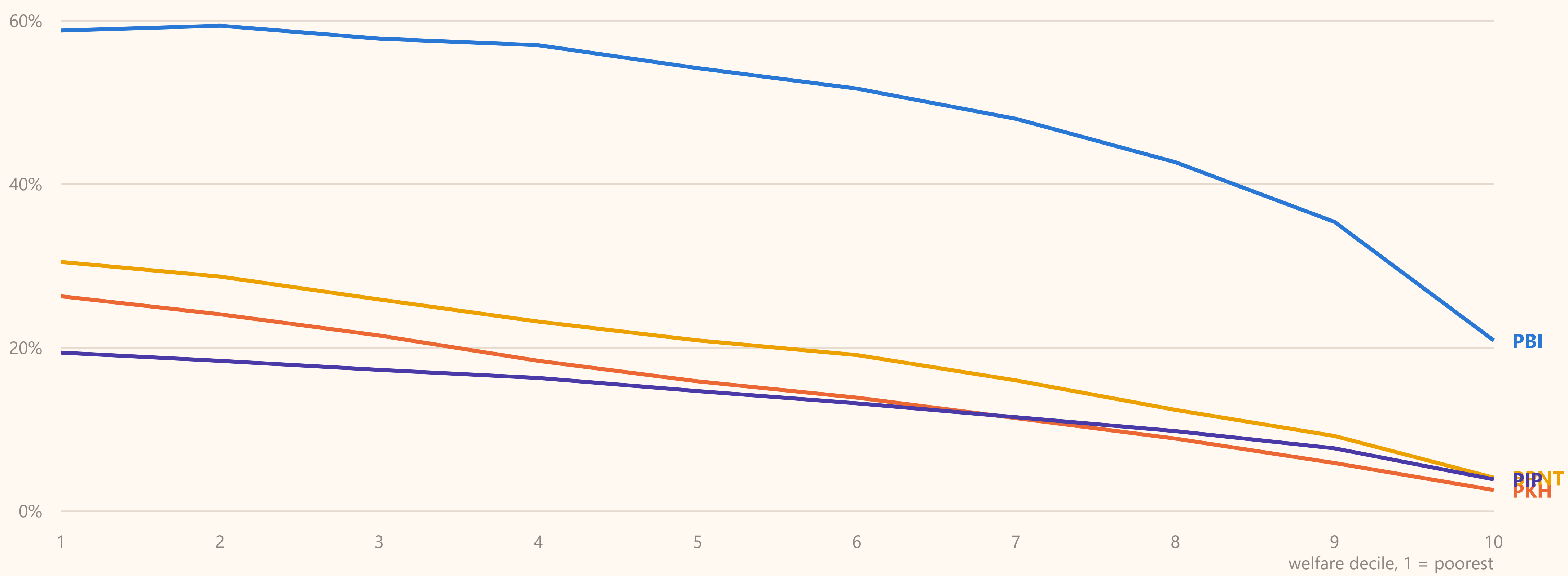
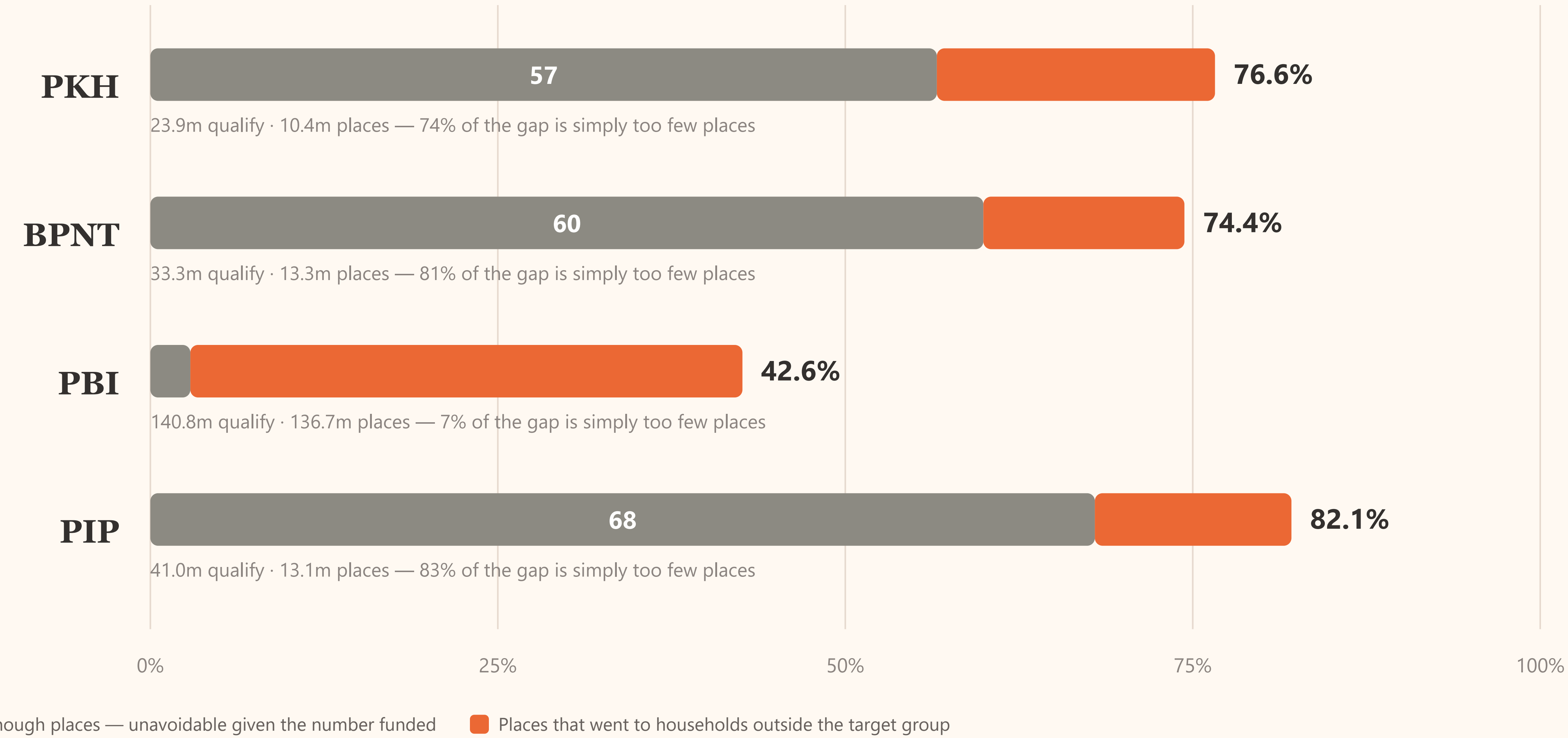


FIGURE 4

How much of the gap is simply too few places

If a programme funds fewer places than there are qualifying households, some of them must be missed however well the targeting works. That floor can be calculated exactly, and it is different for every programme.

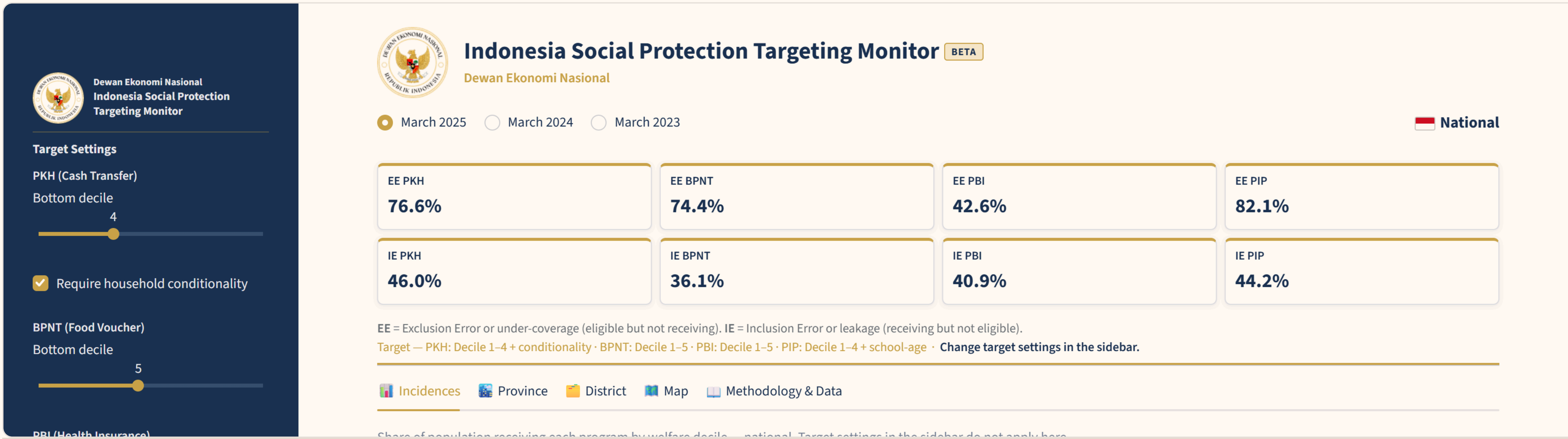


For PKH, BPNT and PIP most of the gap is the number of places, and no data system can change that. **PBI is the exception** — it funds almost as many places as there are people who qualify, so nearly all of its 42.6% is places going to the wrong people. That is the part better data can fix.

THE DATA

These numbers are public and can be checked

The Dewan Ekonomi Nasional targeting monitor computes exclusion and inclusion error directly from Susenas microdata, for three years and for every province and district. Anyone can change the target definition and see the numbers move.



II Is this unusual?

Other countries publish the same measures. What their experience shows is where the error actually comes from.

FIGURE 5

The same range everywhere

Exclusion and inclusion error for social assistance programmes with published figures. Exclusion error runs from about 50% to 90%, inclusion error from about 24% to 70%. Indonesia sits inside both ranges.

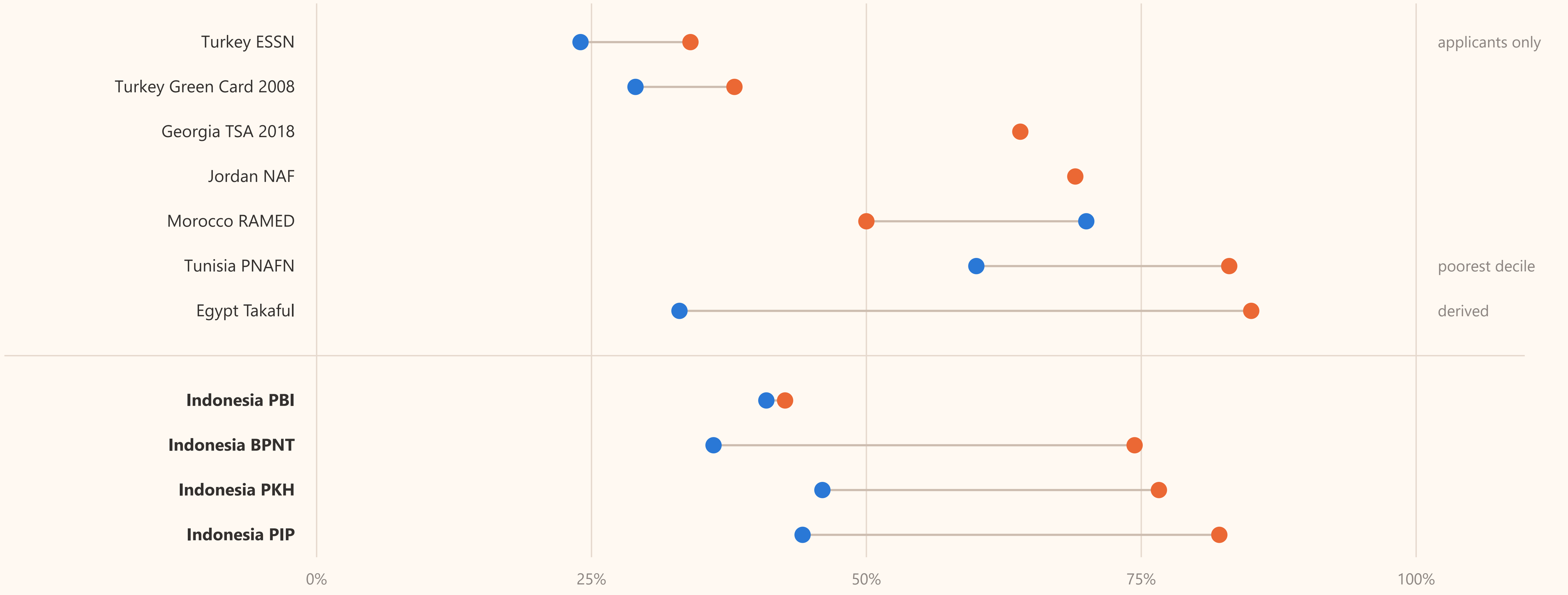
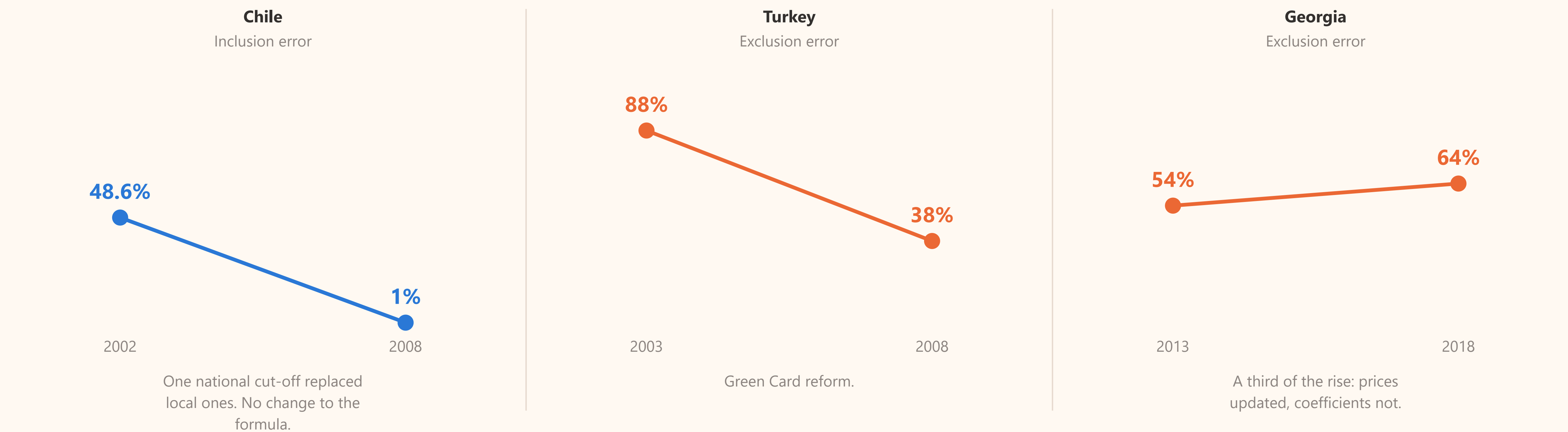


FIGURE 6

What actually changed the numbers elsewhere

In each case where measured error fell sharply, what changed was a rule, a threshold or a process. In none of them was it a better statistical model.



Egypt is the clearest case. Only half of the poorest fifth of households ever applied for Takaful. Researchers calculated that the formula on its own would have admitted 46% non-poor recipients; the figure actually observed was 33%. The programme was more accurate than its formula, because the people who did not need it largely did not apply. The exclusion problem was in **who came forward**, not in the scoring.

III Why is it imperfect?

Two reasons. One is a limit on what any formula can know. The other is everything that happens after the formula.

FIGURE 7

Three ways to decide who is poor

There are three recognised methods. They differ in one thing: how much of a household's income the government can see in its own records.

1

Means test — MT

Measure income directly. Add up income from every source, and test assets, using records the state already holds — tax, payroll, pensions, social security. Nothing is estimated.

Needs: most income visible in records, near-universal formal employment.

Used in: most OECD countries. Chile.

VERIFICATION

2

Proxy means test — PMT

Estimate income from proxies. Collect observable characteristics — housing, assets, household composition, education — and convert them into a score using weights estimated from a survey.

Needs: a household survey and an enumeration round.

Used in: Indonesia, Philippines, Pakistan, Albania, Malawi, Burkina Faso.

ESTIMATION

3

Hybrid means test — HMT

Both at once. Take the income that can be verified from records, estimate only the part that cannot — typically informal earnings — and add them together into one score.

Needs: linked administrative data and a unique ID, plus formal income large enough to be worth verifying.

Used in: Turkey. Common in Eastern Europe.

BOTH

The World Bank does not rank these as better or worse. It presents them as a menu, and says which one fits which situation: a hybrid test **“is recommended when formal income represents a large share of a household’s income”**, and it **“depends on the availability and quality of administrative data”**. That is the criterion that decides what a country can use.

FIGURE 8

Indonesia can verify less than a fifth of income

That is what fixes the method, not the quality of the statistics. Indonesia uses a proxy means test because most income is not visible in any government record.

Means test
Income read directly from records
Needs: Most income visible in records

Chile

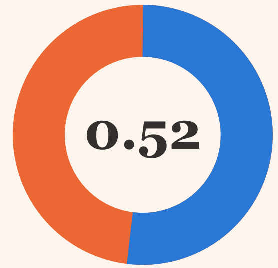
Hybrid means test
Verified income plus estimated informal income
Needs: Tax, payroll and pension records linked

Turkey — the next step

Proxy means test
Welfare estimated from observable proxies
Needs: A household survey

Indonesia is here

Half of what matters is invisible to the formula



52% explained
48% not explained
by the proxies

Indonesia's own PKH proxy-means formulas had an average R^2 of **0.52**. About half the variation in household consumption is explained by the observable proxies. The other half is not.

Why Chile is not the next step

Chile ranks households on recorded income. Recorded income is never higher than true income, so that filter can place a household too low but never too high — it cannot wrongly exclude a poor household. It changes the question from *who is poor?* to *who is clearly not poor?*

That only works when most income is in records. In Indonesia formal employment is under 20% of the workforce. **Turkey is the reachable next step; Chile is the destination.**

FIGURE 9

Five sources of error. Only two are the formula.

Errors arise at two separate stages, and each stage produces both kinds of error. In the outcome data they look identical.

Choosing who to help — pensasaran

Delivering the help — penyaluran

1
Out-of-date register
The household's circumstances changed after it was last recorded.
TARGETING

2
Inaccurate formula
The proxies do not capture the household's real situation.
TARGETING

3
Budget limits
The list is cut to the funded quota, whatever the criteria say.
DELIVERY

4
Business process
Records enter unchecked, appeals go unanswered, payments are delayed.
DELIVERY

5
Elite capture
Local influence decides who appears on the list.
DELIVERY

IV What can digitalisation change?

Registration, verification and payment can be improved now. What the formula can know cannot.

FIGURE 10

Self-registration on its own does not work

No country has succeeded with self-service registration alone. Every system where opening registration improved targeting added four things alongside it.

1

Staff to help people apply

19 of 20 on-demand social registries receive applications at staffed local offices. In Surabaya, **84%** of registrations came through an agent and only 16% were self-registered.

INDONESIA: YES

2

Going out to find people

Brazil ran open registration for five years and reached 22 million families, and still found **over one million** more extremely poor families in a single year once it went looking for them.

INDONESIA: PILOTS ONLY

3

Appeals that get answered

An appeal channel corrects the households the formula missed — but only if the appeals are actually decided.

INDONESIA: PARTLY

4

Checking records before accepting them

Administrative data checked at the moment of application, before the record enters the register.

INDONESIA: NO

Opening registration does help. A randomised trial in 400 Indonesian villages found that when households had to come forward and apply, the people who received benefits were **18 to 19% poorer** than under automatic enrolment, and the very poorest were twice as likely to be reached. But only about **60%** of the very poorest applied — and making the process harder screened out poor and rich equally, so any extra difficulty is a cost borne by the poor for no gain.

FIGURE 11

Indonesia's own audit found both ends open

The state audit board examined the register that preceded DTSEN. Records were being accepted without checking, and appeals were being closed without a decision.

224,635

new applications recorded automatically **without verification** — 15,970 of them without the required photographs. BPK: this creates inclusion error.

7,267

appeals closed automatically without being decided — produces exclusion error. Of these, **735 in Jawa Tengah**, 2,085 in Jawa Barat, 758 in Sumatera Utara.

Turkey's answer is simple

In Turkey's integrated system, when an application contains missing or inconsistent information, an alert appears and **the application is frozen until the information is validated**. The applicant cannot proceed and the record cannot enter the register.

Administrative records behind it are refreshed at least every 45 days, and supply about 40% of the information used to score a household. The remaining 60% still comes from a home visit each year.

This is one transferable design feature, not a whole system: **check the record before you accept it, not after.**

CONCLUSION

What digitalisation can and cannot fix

Of the five sources of error, the two that digitalisation closes fastest are not the statistical formula.

SOURCE OF ERROR	CAN DIGITALISATION FIX IT?	
Out-of-date register	**Yes, directly**	Continuous registration and administrative updates keep the record current.
Business process	**Yes, and fastest**	Check records at the point of application; decide appeals; pay without delay.
Elite capture	**Partly**	An audit trail and published lists make local discretion visible.
Inaccurate formula	**Only up to a limit**	Bounded by how much income can be verified. It improves as formal employment grows, not as the model improves.
Budget limits	**No**	Not a data problem. Better data changes who receives, not how many.

The digital public infrastructure described earlier today is necessary. This is a map of what it will and will not fix.

Appendix

Reference table, for questions.

APPENDIX

Every figure with the population it is measured against

Exclusion and inclusion error can only be compared when you know what each was measured against. These were not measured the same way. The sixth column says how each one differs.

COUNTRY	PROGRAMME	YEAR	EXCLUSION	INCLUSION	MEASURED AGAINST	SOURCE
Turkey	ESSN (refugee cash transfer)	2019	34%	24%	Applicants only — 1.6m of about 3.2m refugees, surveyed before any payment was made. Not comparable with the rest of the table.	Cuevas, Inan, Twose & Çelik (2019), World Bank and WFP
Turkey	Green Card	2003	88%	45%	Poorest decile.	World Bank (2013), Table 3
Turkey	Green Card	2008	38%	29%	Poorest decile. Same programme, five years later.	World Bank (2013), Table 3
Georgia	TSA	2013	54%	—	Bottom quintile of consumption, measured after transfers are removed.	World Bank (2020)
Georgia	TSA	2018	64%	—	As above. About a third of the increase came from prices being updated while the coefficients were not.	World Bank (2020)
Egypt	Takaful	2017	~85%	33%	Exclusion is derived from reported coverage, not stated by IFPRI. Inclusion is measured among beneficiaries, and the population is households with children.	Breisinger et al. (2018), IFPRI
Tunisia	PNAFN	2015	83%	60%	Exclusion against the poorest **decile**; inclusion against the poorest **quintile**. The two are not on the same base.	World Bank (2022)
Tunisia	PNAFN and AMG2	2015	52%	57%	Both against the poorest quintile.	World Bank (2022)
Morocco	RAMED	2019–20	~50%	70%	Exclusion covers RAMED and Tayssir together, against the poorest quintile. Inclusion is the share of beneficiaries outside that quintile.	World Bank (2021)
Jordan	NAF	2010	69%	—	Bottom decile. A second World Bank source reports 83.5% against the poorest quintile — a different denominator, not a contradiction.	World Bank (2012); Silva, Levin & Morgandi (2012)
Chile	Solidario	2002	—	38.6%	Entrants outside the poorest ventile.	Larrañaga & Contreras (2010)
Chile	Solidario	2008	—	1.0%	As above, after local cut-offs were replaced by a single national cut-off. The formula itself did not change.	Larrañaga & Contreras (2010)
Indonesia	PKH	2025	76.6%	46.0%	Deciles 1–4 with programme conditions, households.	Susenas March 2025, own calculation
Indonesia	BPNT	2025	74.4%	36.1%	Deciles 1–5, households.	Susenas March 2025, own calculation
Indonesia	PBI	2025	42.6%	40.9%	Deciles 1–5, individuals.	Susenas March 2025, own calculation
Indonesia	PIP	2025	82.1%	44.2%	Deciles 1–4, school age 6–27, individuals.	Susenas March 2025, own calculation

Sources. Susenas March 2023, 2024 and 2025, own calculation following the definitions of the DEN Indonesia Social Protection Targeting Monitor. Cross-country figures from World Bank and IFPRI publications, each with its own definition of the target population — see the note under each figure. Framework and illustrations from *Salah Sasaran* (Mei 2026). Audit findings from BPK RI, IHPS II 2024.